FLIGHT RESERVATION SYSTEM

Human Computer interaction

OVERVIEW

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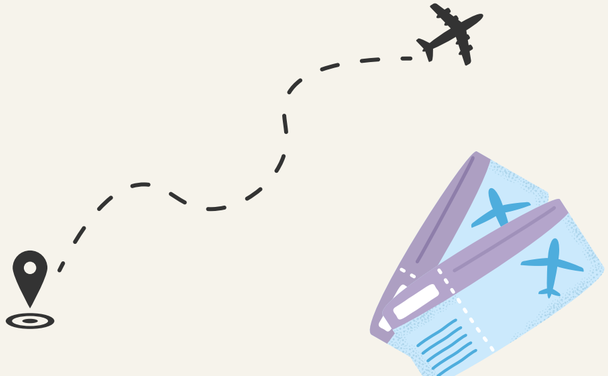
INTRODUCTION

The airline reservation systems one of the important systems to a lot of people who travel using an airplane, is a digital platform used by airlines to manage flight bookings and passenger information. It enables travellers to search for available flights, reserve seats, and purchase tickets. The system provides the booking process and organizing flight schedules and seat allocations.

Proposal:

Flight reservation problems are common, but we will provide effective solutions that satisfy our customers of the solutions we will provide will contact customers quickly in the event of flight delays or cancellations, and provide compensation for any problem or inconvenience we will process refunds within a maximum period of two days and by improving airline reservation systems we aim to simplify booking processes, provide real-time information and enhance the overall travel experience for customers Predictive analytics can also help airlines adjust prices based on demand and customize offers to different customer segments It is also recommended to contact the airline directly for information about cancellation policies and options.



The User:

Primary user:

Ordinary travellers and business travellers

System users are often over the age of 18

They are looking for an easy and fast booking experience

They want to find the best offers and prices

Secondary user:

Travel managers in companies and travel agencies

They manage flight bookings for employees or customers

Some of the tasks:

-Search for flights based on dates and locations.

-The system allows users to book flight tickets through a variety of airline companies at different prices.

-Access flight information such as departure and arrival times and duration.

-Receive reminders and alerts about flight changes or announcements of special offers.

Interviews, Field study or Questionnaire results:

- features and improvements over existing systems?

- **What are your opinion about airline travel and the process of booking flights?**
- **What are some challenges you typically encounter when making airline reservations?**
- **What would you like to improve in the program? What can the program do to attract your attention?**

Interviews, Field study or Questionnaire results:

In a comprehensive exploration of airline travel preferences, both Qualitative, interviews and quantitative questionnaires provided illuminating insights into the experiences and opinions of travellers.

Participants revealed a range of travel frequencies, from frequent flyers to occasional travellers, .highlighting the diverse nature of air travel habits

Moreover, the overall sentiment towards airline travel and the booking process was predominantly positive, underscoring the convenience and utility it offers in modern travel. Despite the overall positivity, participants also candidly shared the challenges they encounter when making airline reservations

Interviews, Field study or Questionnaire results:

Key priorities included user-friendly interfaces, efficient booking processes, and seamless management of bookings Additionally, reliable customer support and access to comprehensive flight information were deemed essential by many.

Participants offered valuable suggestions for improving flight booking systems, ranging from reducing waiting times between flights to streamlining application usage.

Requirements:

Functional Requirements:

1- Effective Search: Users should be able to search for flights based on criteria such as timing, destination.

2- Comprehensive Information Presentation: The system must display to user detailed information about flights, including prices, duration, and availability.

3- Easy Booking Processes: Users should be able to book flights and complete payment processes smoothly, with the option to store passenger information for future use.

Requirements:

Non-Functional Requirements:

1- Responsive Design: The interface should be responsive across various devices and screen sizes to provide a flexible user experience.

2- Multi-language Support: The interface should support multiple languages to ensure accessibility for users worldwide.

3- Secure Payment Integration: Payment processing must be secure and compliant with industry standards.

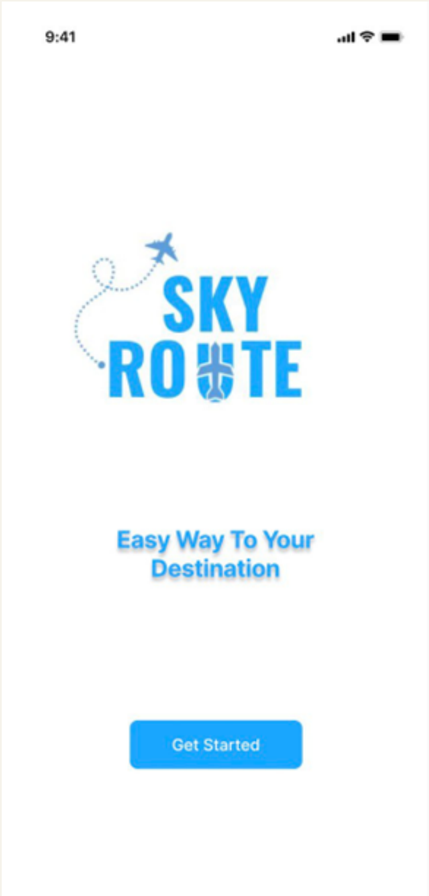
LOGO

Our logo has the name of our system's name "sky route" we choose two shades of the colour blue, also we used two shapes of a plane

we wanted to make it obvious to users that is refers to a flight reservation system so the users can have recognisable definition about it, we wanted to keep it simple and clear as possible.

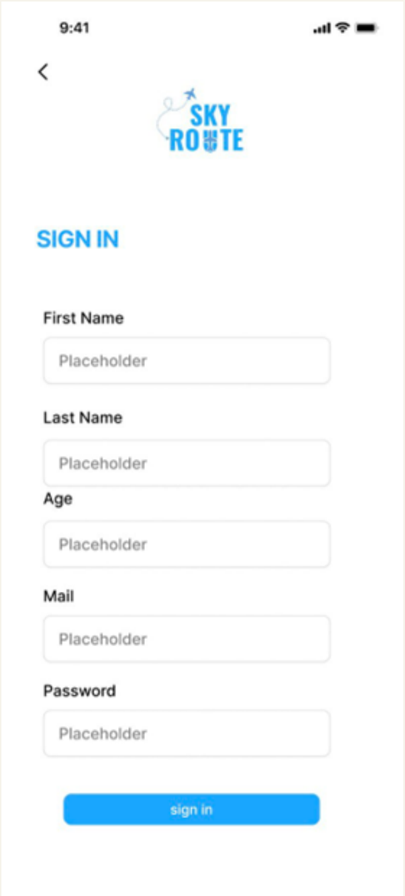
PROTOTYPE

1- START



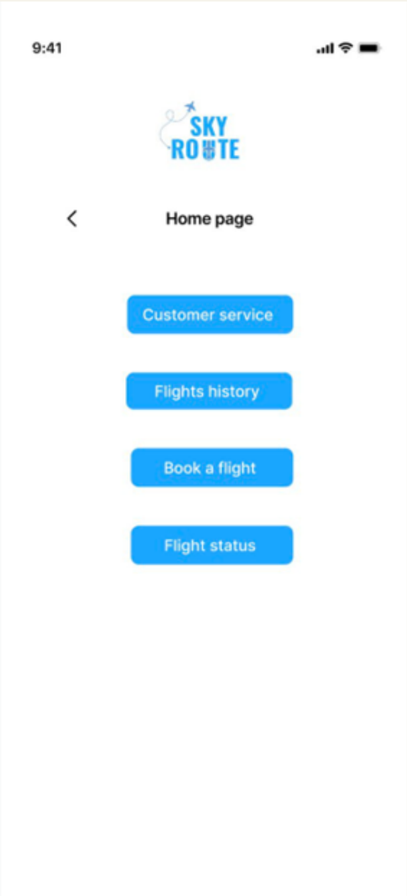
Contains the logo and the first bag appears to the user

2- SIGN UP



When the user click get started this page will appear to enter the user information in order to book a ticket

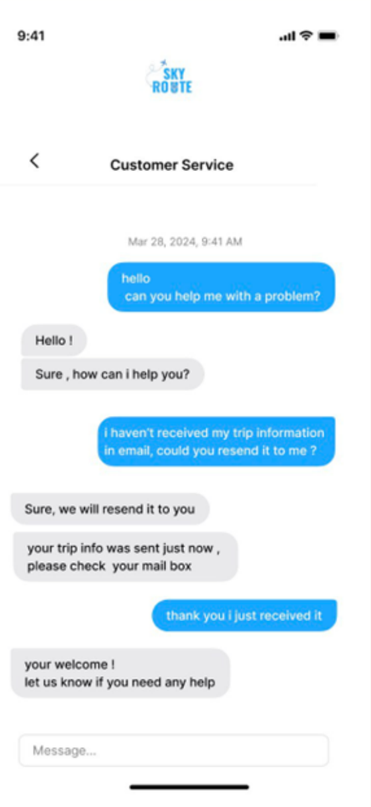
3-HOME PAGE



After the user sign up the home page will apper and the user can chose from the service provided

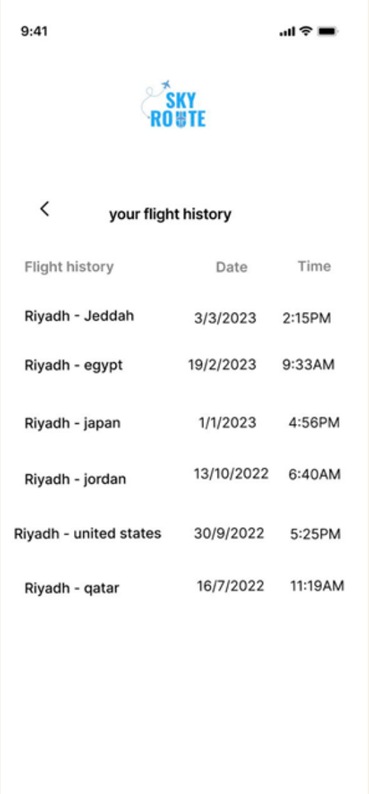
PROTOTYPE

4- CUSTOMER SERVICE



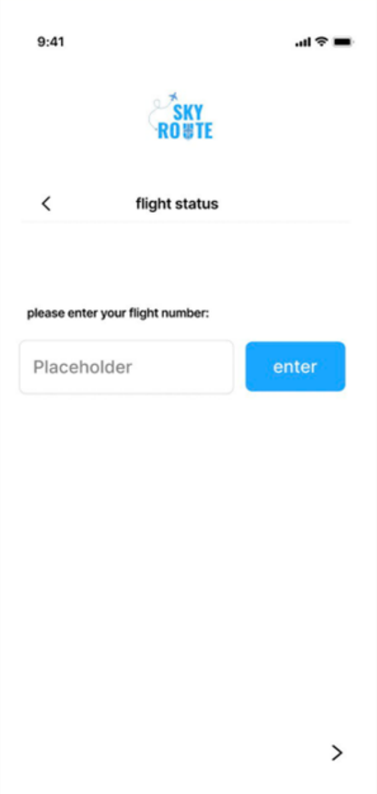
This page will appear after the user chose the customer service from the home page , customer service help the users if they have A problem, question, or even an opinion

5- FLIGHT HISTORY



This page will appear after the user chose your flight history from the home page. This page will display previous flights, their dates and times.

6- FLIGHT STATUS



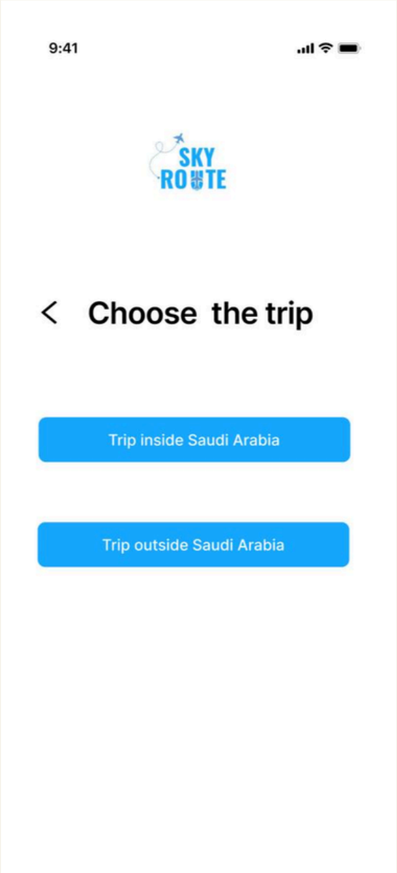
This page will appear after the user chose the flight status from the home page .On this page, you enter the flight number that you want to know more information about

PROTOTYPE

7- FLIGHT SATUS

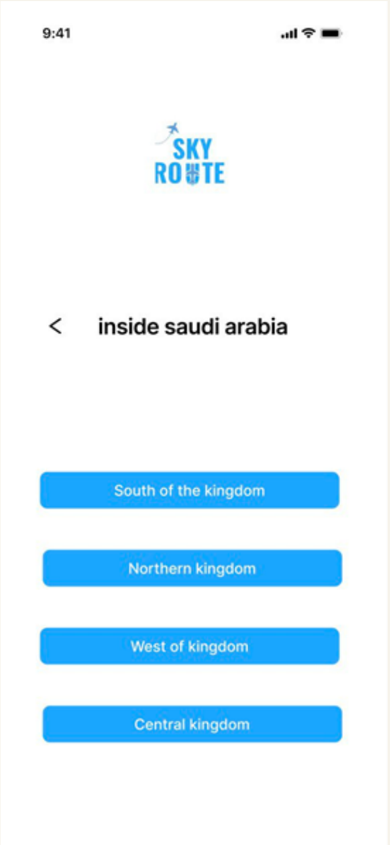
After you click enter the flight information will appear

8- CHOOSE TRIP



This page will appear after the user chose book a flight from the home page This page asks if you want to book a ticket to a place inside Saudi Arabia or a place outside Saudi Arabia Make it easier for you to choose if you are hesitant

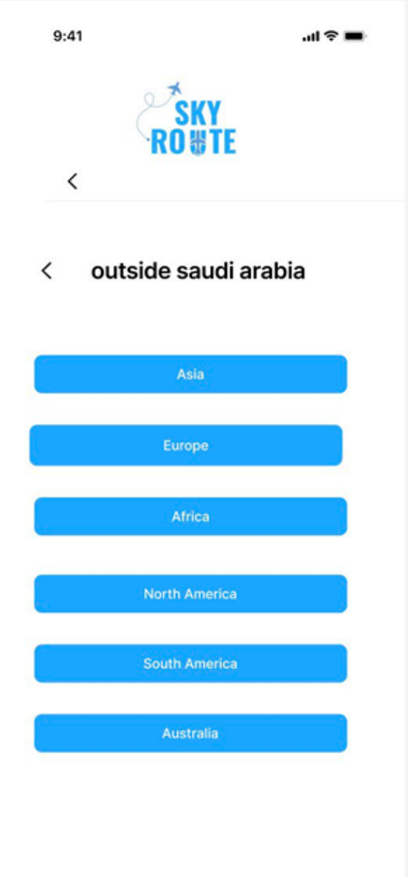
9- CHOOSE TRIP



After the user click inside Saudi ArabiaYou can choose north. South. West. Eastern Saudi Arabia

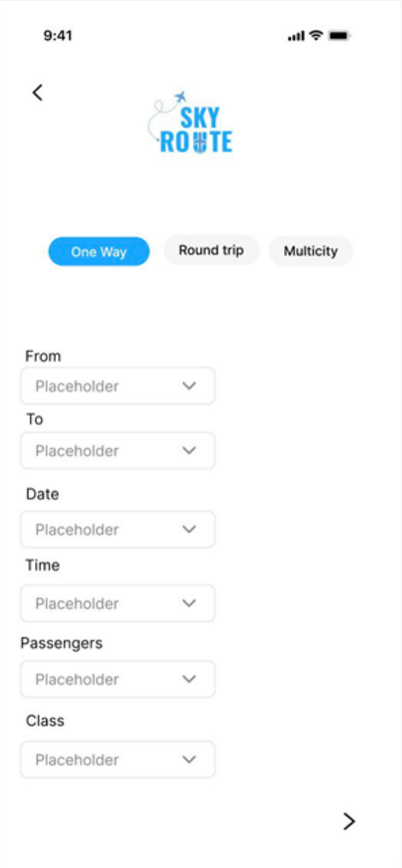
PROTOTYPE

10- CHOOSE TRIP



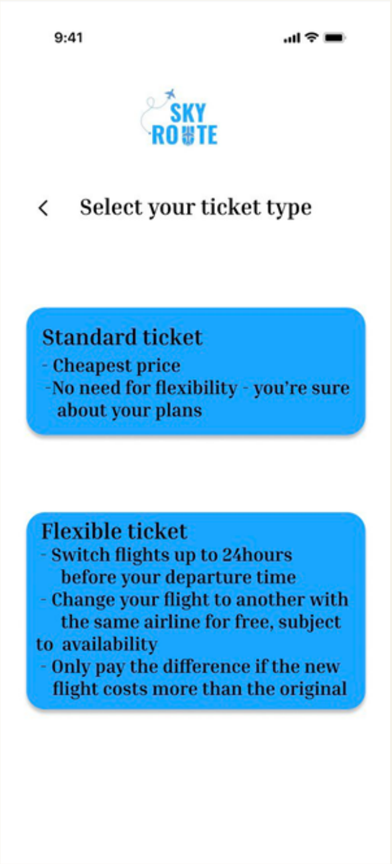
Saudi Arabia this page will give the user a lot of chooses

11- TRIP INFO



On this page, the user can determine if he wants one way or round trip or multicity In addition, he specifies which country he will fly from to which country he will go to, the dates and time, the number of passengers and the type of ticket

12- TICKET TYPE



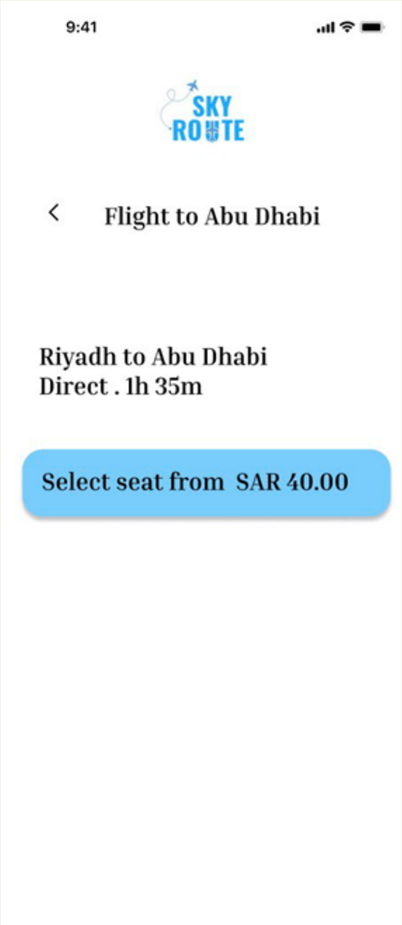
After all the required information for a trip has been entered by the user, this page will appear On this page, the user can determine the type of ticket according to his budget and needs

PROTOTYPE

13- LUGGAGE

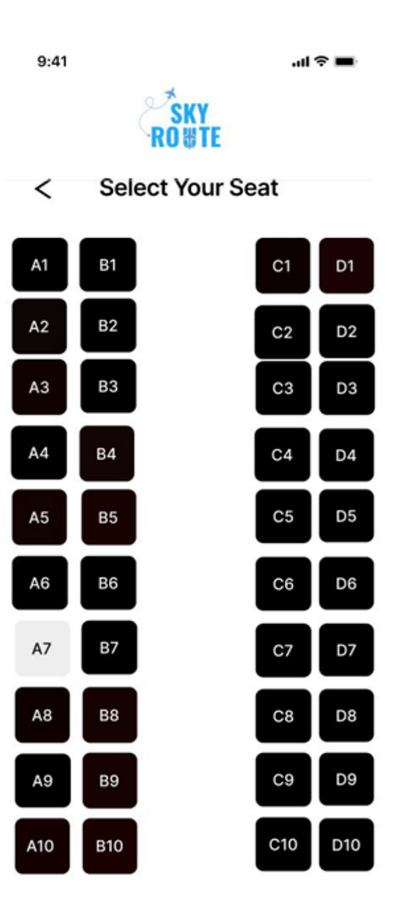
On this page, the user can choose either 1 carry-on bag or 1 checked bag , depending on the baggage

14- SELECT SEAT



In this page , the user can reserve a specific seat on the plane starting from 40 riyals, depending on the seat

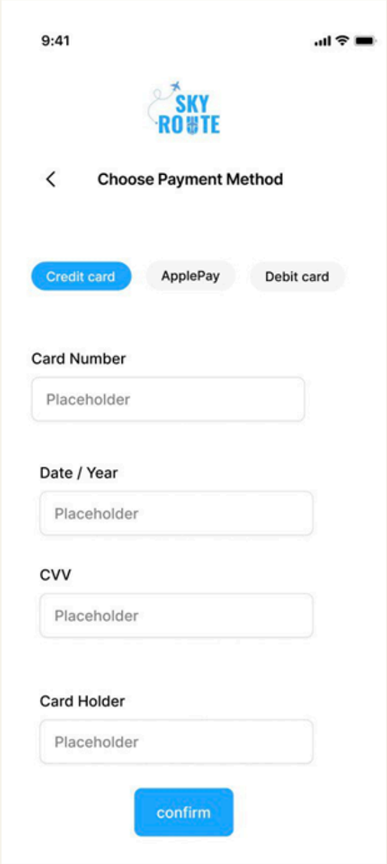
15- CHOOSE SEAT



On this page, the user can view the seats as if he were inside the plane and choose the seat he wants

PROTOTYPE

16- PAYMENT



On this page, the user can choose the method that suits him to pay and enter all the information related to the credit card

17- GET BOARDING BASS

This page appears after the user completes payment

18- BOARDING BASS

Boarding Pass: It shows flight information, gate information, take-off and landing times And the seat that was chosen

TASKS

MOST IMPORTANT AND FREQUENT TASKS FOR THE AIRLINE RESERVATION SYSTEM:

1. FLIGHT SEARCH: USERS CAN LOOK FOR AVAILABLE FLIGHTS BASED ON THEIR TRAVEL DATES AND DESTINATIONS.

2. SEAT RESERVATION: USERS CAN CHOOSE AND BOOK SEATS ACCORDING TO THEIR PREFERENCES.

3. TICKET PURCHASE: USERS CAN SECURELY BUY TICKETS FOR THEIR CHOSEN FLIGHTS.

4. FLIGHT SCHEDULE MANAGEMENT: THE SYSTEM MANAGES AND DISPLAYS ACCURATE FLIGHT SCHEDULES.

5. PASSENGER INFORMATION MANAGEMENT: USERS INPUT AND MANAGE PERSONAL DETAILS.

DESIGN DESCRIPTION

OUR DESIGN IS STRAIGHTFORWARD AND USER-FRIENDLY. HERE'S HOW WE APPROACHED IT:

1. GRAPHICAL USER INTERFACE (GUI): WE DESIGNED A SIMPLE INTERFACE WITH EASY-TO-UNDERSTAND ICONS AND BUTTONS FOR USERS.

2. FORM-BASED INTERACTION: WE USED FORMS FOR USERS TO INPUT THEIR INFORMATION.

3. MENU-DRIVEN NAVIGATION: CLEAR MENU OPTIONS GUIDE USERS THROUGH THE BOOKING PROCESS.

4. FEEDBACK MECHANISMS: OUR DESIGN PROVIDES INSTANT FEEDBACK TO USERS.

5. RESPONSIVE DESIGN: THE DESIGN ADAPTS TO DIFFERENT DEVICES AND SCREEN SIZES.

DESIGN DESCRIPTION

WE BELIEVE OUR DESIGN WILL IMPROVE THE USER EXPERIENCE BY:

- STREAMLINING BOOKING PROCESS: THE SIMPLE INTERFACE MAKE IT EASY FOR USERS TO FIND FLIGHTS.

- ENHANCING ENGAGEMENT: INTERACTIVE ELEMENTS AND INSTANT FEEDBACK KEEP USERS ENGAGED AND INFORMED.

- INCREASING ACCESSIBILITY: THE RESPONSIVE DESIGN ENSURES USERS CAN ACCESS THE SYSTEM FROM ANY DEVICE.

- ENSURING ACCURACY: ACCURATE FLIGHT SCHEDULES AND SEAT ALLOCATIONS PROVIDED BY THE SYSTEM REDUCE ERRORS.

TESTING

No.	Task Name	Task Time	Number of errors	Comments	Suggestion to improve task design
1	Sign up	2 min	0	task done successfully	none
2	Book ticket	5 min	0	successfully with no issues	none
3	Choose seat	7 min	1	user interface confusion	add more information about if the seat was in front of the plane or not
4	Select ticket type	1 min	0	task done successfully	none
5	Select luggage	2 min	0	task done successfully	none
6	Payment	6 min	0	task done successfully	none
7	Search flight status	6 min	0	Done successfully with no issues	none
8	Show flight history	1 min	0	task done successfully.	none
9	Choose trip inside or outside Saudi Arabia	2 min	0	Done successfully with no issues	none
10	Fill trip info	10 min	0	task done successfully	none

FROM THE DATA PROVIDED, IT'S CLEAR THAT THE TASK DESIGN GENERALLY LEADS TO SUCCESSFUL COMPLETION, BUT THERE ARE AREAS FOR IMPROVEMENT. SPECIFICALLY, TASK 3, "CHOOSE SEAT," HAD SOME USER INTERFACE CONFUSION, RESULTING IN ONE ERROR. THEREFORE, TO ENHANCE TASK DESIGN, IT'S SUGGESTED TO ADD MORE INFORMATION ABOUT THE SEAT LOCATION, SUCH AS WHETHER IT IS IN FRONT OF THE PLANE OR NOT. OVERALL, THE TASKS WERE COMPLETED WITH A TOTAL OF ONLY ONE ERROR OUT OF TEN, INDICATING A FAIRLY SUCCESSFUL DESIGN.

Background information & Related work:

There are many applications that serve the same purpose as our system. Compared to the Booking application, our application is characterized by its speed in responding to commands, so that in the Booking application, when moving to another page, it takes longer than the time it takes our application to move between pages. Also, in the booking application, every time you search for a specific flight, the prices vary by a small and unnoticeable percentage.

Conclusion and Future work :

IN CONCLUSION, AVIATION PROBLEMS ARE CONSIDERED ONE OF THE VERY COMMON PROBLEMS ,AND ANNOYING TO SOME CUSTOMERS

BUT WE HAVE PROVIDED SOME SOLUTIONS THAT CUSTOMERS ARE INTERESTED IN WE HAVE DEVELOPED AND IMPROVED SOME SERVICES AND PROVIDED BETTER OPTIONS FOR TRAVELERS IN THE FUTURE, WE WILL DEVELOP THE SERVICE OF PROVIDING AN INTELLIGENT SYSTEM THAT SUGGESTS SUITABLE OFFERS AND OPTIONS TO TRAVELERS BASED ON THEIR PREFERENCES AND PREVIOUS BOOKING HISTORY

WE WILL USE SOME MODERN TECHNOLOGIES AND PREDICTIVE ANALYTICS TO DEVELOP THIS SERVICE IN THE FUTURE BECAUSE WE CARE ABOUT THE CUSTOMER'S SATISFACTION, COMFORT, AND ENJOYMENT OF THE TRIP.

References :

1. EXPEDIA [ONLINE]. AVAILABLE: HTTPS://WWW.EXPEDIA.COM/

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THANK YOU